مجال علوم الحاسبات
VIII. FIELD OF COMPUTER SCIENCE

البرامج الدراسية
Study Programs

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Computer Science Program — برنامج علوم الحاسب

III. Software Engineering — III. هندسة البرمجيات

Study Plan — الخطة الدراسية

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
1	MAT111	Mathematics I	3	0	1	0	4	3	150	6
	PHY211	Physics II	2	2	0	0	4	3	150	6
	CSE014	Structured Programming	2	3	0	0	5	3	180	6
	UC1	University Requirement (1)	2	0	0	0	2	2	90	4
	UC2	University Requirement (2)	2	0	0	0	2	2	90	4
	UE1	Elective University (1)	2	0	0	0	2	2	90	4
							19	**15**	**750**	**30**
2	MAT112	Mathematics II	3	0	1	0	4	3	150	6
	MAT131	Probability & Statistics I	2	1	2	0	5	3	195	6
	CSE015	Object Oriented Programming	2	3	0	0	5	3	180	6
	CSE315	Discrete Mathematics	2	0	2	0	4	3	150	6
	UC3	University Requirement (3)	2	0	0	0	2	2	90	4
	UE2	Elective University (2)	2	0	0	0	2	2	90	4
							22	**16**	**855**	**32**

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Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
3	MAT212	Linear Algebra	2	2	0	0	4	3	150	6
	MAT231	Probability & Statistics II	2	2	1	0	5	3	195	6
	CSE111	Data Structures	2	3	0	0	5	3	180	6
	CSE131	Logic Design	2	2	2	0	6	3	195	6
	AIE111	Artificial Intelligence	2	2	1	0	5	3	180	6
	UC4	University Requirement (4)	2	0	0	0	2	2	90	4
							27	**17**	**990**	**34**
4	MAT312	Differential Equations	2	0	2	0	4	3	150	6
	CSE132	Computer Architecture & Organization	2	2	1	0	5	3	165	6
	CSE211	Web Programming	2	2	1	0	5	3	165	6
	CSE221	Database Systems	2	2	1	0	5	3	180	6
	CSE191	Field Training 1 in Computer Science	2	0	0	0	2	2	60	4
	CSE251	Software Engineering	2	2	1	0	5	3	180	6
							26	**17**	**900**	**34**
5	CSE233	Operating Systems	2	2	1	0	5	3	165	6
	E1	Elective Course (1)	2	2	1	0	5	3	180	6
	CSE261	Computer Networks	2	2	1	0	5	3	165	6
	AIE121	Machine Learning	2	2	1	0	5	3	180	6
	CSE352	Systems Analysis & Design	2	2	1	0	5	3	165	6
	UC5	University Requirement (5)	2	0	0	0	2	2	90	4
							27	**17**	**945**	**34**
	CSE112	Design & Analysis of Algorithms	3	0	3	0	6	4	195	8

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Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
6	CSE241	Security of Information Systems	2	2	1	0	5	3	150	6
	CSE323	Advanced Database Systems	2	2	1	0	5	3	180	6
	E2	Elective Course (2)	2	2	1	0	5	3	180	6
	CSE292	Field Training 2 in Computer Science	2	0	0	0	2	2	60	4
	UC6	University Requirement (6)	2	0	0	0	2	2	90	4
							25	**17**	**855**	**34**
7	CSE363	Cloud Computing	2	2	1	0	5	3	180	6
	CSE454	Advanced Software Engineering	2	2	1	0	5	3	180	6
	AIE323	Data Mining	2	2	1	0	5	3	180	6
	CSE493	Graduation Project 1	2	3	0	0	5	3	180	6
	UC7	University Requirement (7)	2	0	0	0	2	2	90	4
	E3	Elective Course (3)	2	2	1	0	5	3	180	6
							27	**17**	**990**	**34**
8	CSE411	Theory of Computation & Compilers	2	2	1	0	5	3	180	6
	CSE474	Distributed Information Systems	2	2	1	0	5	3	180	6
	CSE494	Graduation Project 2	2	3	0	0	5	3	180	6
	E4	Elective Course (4)	2	2	1	0	5	3	180	6
	E5	Elective Course (5)	2	2	1	0	5	3	180	6
	UE3	Elective University (3)	2	0	0	0	2	2	90	4
							27	**17**	**990**	**34**
							200	**133**	**7305**	**266**

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Alamein- CS PROGRAM/Big data Track- COURSE MAP 2020 - 127 CR HRS



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Elective (E1)

CSE234	Embedded Systems
CSE271	Introduction to Parallel Computing
CSE281	Image Processing
CSE313	Mobile Programming

Elective (E2)

CSE322	Big Data Analytics 1
AIE212	Knowledge-Based Systems
AIE231	Neural Networks
AIE241	Natural Language Processing

Elective (E3)

CSE312	Advanced Web Programming
CSE464	Internet of things
AIE343	Machine Learning for Text Mining
AIE424	Intelligent Decision Support Systems
AIE425	Intelligent Recommender Systems

Elective (E4)

CSE426	Selected Topics in Data Science
CSE424	Data Warehousing
CSE453	Building Information Systems
CSE455	Selected Topics in Software Engineering

Elective (E5)

AIE332	Deep Learning
AIE342	Advanced Methods for Data Analysis
AIE314	AI-Based Programming
AIE322	Advanced Machine Learning

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برنامج علوم الذكاء الاصطناعي

Artificial Intelligence Science Program

Program Description

وصف البرنامج

– The program grants the deep knowledge and skills needed to transform big amounts of data into executable decisions. The program focuses on how to use complex inputs like vision, languages, and big databases in decision-making and enhance human power.

– يمنح البرنامج المعرفة والمهارات العميقة المطلوبة لتحويل كميات كبيرة من البيانات إلى قرارات قابلة للتنفيذ. يركز البرنامج ومناهجه على كيفية استخدام المدخلات المعقدة مثل الرؤية واللغة وقواعد البيانات الضخمة في اتخاذ القرارات أو تعزيز القدرات البشرية.

Graduate attributes

مواصفات الخريج

In addition, the graduate of the Artificial Intelligence Science program has all the modules of graduates of computer science programs. The graduates have the following characteristics intensively.

– The ability to build software and intelligent agent systems.
– The ability to recognize natural language systems and their implementation.

بالإضافة الى تمتع خريج برنامج علوم الذكاء الاصطناعي بكافة مقومات خريجي برامج علوم الحاسب، يتسم الخريجين بالسمات التالية بشكل مكثف.

– القدرة على إنشاء برمجيات وأنظمة العميل الذكي.
– القدرة على أنظمة التعرف على اللغات الطبيعية وبرمجتهما.

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- The ability to build machine-learning systems, implement them, and measure their performance.

- القدرة على إنشاء أنظمة تعلم الآلة وبرمجتها وقياس أدائها.

Job Opportunities

مجالات عمل الخريج

- International companies working in natural languages processing – Smart programming companies – Smart home components manufacturers – digital transformation companies' startups

- الشركات الدولية العاملة في معالجة اللغات الطبيعية – شركات البرمجة الذكية – شركات تصنيع منتجات المنازل الذكية – شركات التحول الرقمي الناشئة.

Benchmark

المرجعية

- The program has adopted the general knowledge and skills issued from the National authority for quality assurance and accreditation of education (NARS) for computer science graduate which can be found through the following link. http://naqaae.eg/wp-content/uploads/2014/PDF/26.pdf
- The bylaws of computer science faculties have been studied and referenced in the following universities: Stanford University, Carnegie Mellon University, Nile University. https://www.cs.cmu.edu/bs-in-artificial intelligence/curriculum
- Computer Reference Standard https://www.abet.org/accreditation/accreditation-criteria/criteria-for-accrediting-computing-programs-2018-2019

- وقد تبنى البرنامج المعارف والمهارات الصادرة من الهيئة القومية لضمان جودة التعليم والاعتماد (NARS) لخريجي الحاسبات والمعلومات والتي يمكن الاطلاع عليها من خلال الرابط التالي: http://naqaae.eg/wp-content/uploads/2014/PDF/26.pdf
- تم الاطلاع والرجوع للوائح كليات الحاسبات في جامعة ستانفورد - كارنيجي ميلون – جامعة النيل.
- المعايير المرجعية للحوسبة
- https://www.abet.org/accreditation/accreditation--criteria/criteria

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Artificial Intelligence Science Program

برنامج علوم الذكاء الاصطناعي

Study Plan

الخطة الدراسية

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
1	MAT111	Mathematics I	3	0	1	0	4	3	150	6
	PHY211	Physics II	2	2	0	0	4	3	150	6
	CSE014	Structured Programming	2	3	0	0	5	3	180	6
	UC1	University Requirement (1)	2	0	0	0	2	2	90	4
	UC2	University Requirement (2)	2	0	0	0	2	2	90	4
	UE1	Elective University (1)	2	0	0	0	2	2	90	4
							19	**15**	**750**	**30**
2	MAT112	Mathematics II	3	0	1	0	4	3	150	6
	MAT131	Probability & Statistics I	2	1	2	0	5	3	195	6
	CSE015	Object Oriented Programming	2	3	0	0	5	3	180	6
	CSE315	Discrete Mathematics	2	0	2	0	4	3	150	6
	UC3	University Requirement (3)	2	0	0	0	2	2	90	4
	UE2	Elective University (2)	2	0	0	0	2	2	90	4
							22	**16**	**855**	**32**
	MAT212	Linear Algebra	2	2	0	0	4	3	150	6

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Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
3	MAT231	Probability & Statistics II	2	2	1	0	5	3	195	6
	CSE111	Data Structures	2	3	0	0	5	3	180	6
	CSE131	Logic Design	2	2	2	0	6	3	195	6
	AIE111	Artificial Intelligence	2	2	1	0	5	3	180	6
	UC4	University Requirement (4)	2	0	0	0	2	2	90	4
							27	**17**	**990**	**34**
4	MAT312	Differential Equations	2	0	2	0	4	3	150	6
	CSE132	Computer Architecture & Organization	2	2	1	0	5	3	165	6
	CSE221	Database Systems	2	2	1	0	5	3	180	6
	CSE281	Image Processing	2	2	1	0	5	3	180	6
	AIE121	Machine Learning	2	2	1	0	5	3	180	6
	AIE191	Field Training 1 in AI Science	2	0	0	0	2	2	60	4
							26	**17**	**915**	**34**
5	CSE233	Operating Systems	2	2	1	0	5	3	165	6
	CSE251	Software Engineering	2	2	1	0	5	3	180	6
	CSE261	Computer Networks	2	2	1	0	5	3	165	6
	AIE231	Neural Networks	2	2	1	0	5	3	180	6
	AIE323	Data Mining	2	2	1	0	5	3	180	6
	UC5	University Requirement (5)	2	0	0	0	2	2	90	4
							27	**17**	**960**	**34**
6	CSE112	Design & Analysis of Algorithms	3	0	3	0	6	4	195	8
	AIE212	Knowledge-Based Systems	2	2	1	0	5	3	180	6

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Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	AIE213	Optimization Techniques	3	0	3	0	6	3	180	6
	AIE241	Natural Language Processing	2	2	1	0	5	3	180	6
	AIE292	Field Training 2 in AI Science	2	0	0	0	2	2	60	4
	E1	Elective Course (1)	2	2	1	0	5	3	180	6
							29	**18**	**975**	**36**
7	CSE363	Cloud Computing	2	2	1	0	5	3	180	6
	AIE322	Advanced Machine Learning	2	2	1	0	5	3	180	6
	AIE332	Deep Learning	2	2	1	0	5	3	180	6
	AIE493	Graduation Project 1	2	3	0	0	5	3	180	6
	E2	Elective Course (2)	2	2	1	0	5	3	180	6
	UC6	University Requirement (6)	2	0	0	0	2	2	90	4
							27	**17**	**990**	**34**
8	AIE425	Intelligent Recommender Systems	2	2	1	0	5	3	180	6
	AIE314	AI-Based Programming	2	2	1	0	5	3	180	6
	AIE494	Graduation Project 2	2	3	0	0	5	3	180	6
	E3	Elective Course (3)	2	2	1	0	5	3	180	6
	UC7	University Requirement (7)	2	0	0	0	2	2	90	4
	UE3	Elective University (3)	2	0	0	0	2	2	90	4
							24	**16**	**900**	**32**
							201	**133**	**7335**	**266**

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Alamein- AIS PROGRAM- COURSE MAP 2020 - 133 CR HRS



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Elective (E1)

CSE211	Web Programming
CSE234	Embedded Systems
CSE271	Introduction to Parallel Computing
CSE322	Big Data Analytics 1
CSE344	Introduction to Cyber Security
CSE382	Computer Graphics
CSE383	Computer Vision
CSE411	Theory of Computation & Compilers

Elective (E2)

CSE464	Internet of things
CSE475	High Performance Computing
CSE485	Computer Games Theory
AIE417	Selected Topics in Artificial Intelligence 1
AIE315	Computational Logic
AIE316	Evolutionary Algorithms
AIE342	Advanced Methods for Data Analysis
AIE343	Machine Learning for Text Mining
AIE351	Robotics Design

Elective (E3)

CSE487	Mixed & Augmented Reality
CSE488	Visualization & Animation
AIE418	Selected Topics in Artificial Intelligence 2
AIE424	Intelligent Decision Support Systems
AIE426	Decision Making Under Uncertainty
AIE427	Statistical Pattern Recognition
AIE444	Question Answering Systems
AIE457	Mobile Robot Development

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